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1. Introduction

PopBeauty is a fictional beauty retailer operating 13 stores across 7 UK regions. It is valued at approximately £4.2 billion and characterised by strong brand loyalty and sensitivity to product freshness (Statista, 2024). Its portfolio spans over 65 brands across 6 categories: Makeup, Skincare, Fragrance, Haircare, Beauty Tools, and Beauty Accessories. The network includes 1 Flagship, 5 Large stores, 4 Medium stores, and 3 Boutiques, supported by 6 regional warehouses across the UK. This project designs a relational database for PopBeauty, implements it in SQLite generates synthetic data, and uses SQL to identify business insights. The main business problem is inventory imbalance; some stores are overstocked whereas higher-demand locations face shortages that slash revenue.

2. Database Design and Implementation

2.1. Conceptual Design and Entity-Relationship Model

Following Coronel and Morris (2019), the design process began by defining the mini-world for PopBeauty, defining the entities Stores, Warehouses, Brands, Categories, Products, Store_Inventory, Warehouse_Inventory, Sales, and Shipments. The entity-relationship (ER) diagram is presented as follows, and summarised in Appendix A.

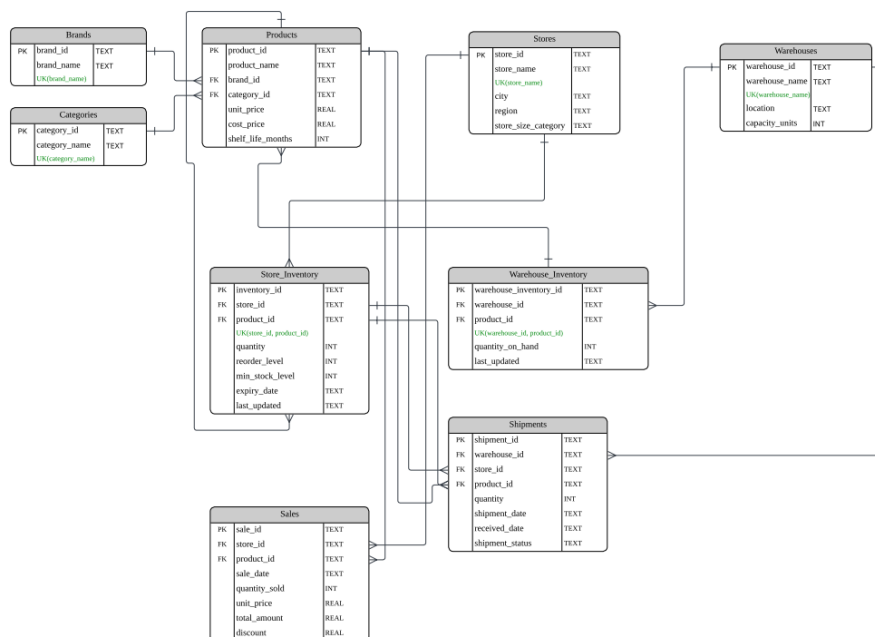


Figure 1: ER Diagram for PopBeauty Database

2.2. *Normalisation*

The schema is normalised to Third Normal Form (3NF) to remove repeating groups, partial dependencies, and transitive dependencies. For example, brand name and category name are held in their own entities and referenced via foreign keys rather than repeated on every product record.

2.3. *Synthetic Data Generation*

Synthetic data is generated using Python and ChatGPT, guided by the schema and UK prestige beauty retail context. All outputs were validated for consistency. The main entities Store_Inventory, Warehouse_Inventory, Sales, and Shipments each contain over 1,000 records.

3. **The Business Problem**

PopBeauty's 13 stores do not receive inventory in proportion to where demand is strongest. The data reveals a consistent and commercially costly pattern of smaller stores accumulating stock they cannot convert into sales, while higher-footfall locations run short of the products customers want. This is not a conventional supply problem. The stock exists within the network. Chopra and Meindl (2016) describe this type of 'mismatch between supply and demand location' as a leading cause of both stockout costs and excess holding costs in retail supply chains.

PopBeauty's inventory is not aligned with demand. Smaller stores accumulate stock they cannot sell, while higher-footfall stores face shortages. This is an allocation problem rather than a supply shortage, creating both stockout and holding costs.

The scale of the imbalance emerges clearly from a comparison of units on hand against units sold, as shown in Table 1.

Store	Size	Units on Hand	Units Sold	Sell-Through Ratio
PopBeauty London	Flagship	4,075	1,746	0.43
PopBeauty Glasgow	Large	4,832	767	0.16
PopBeauty Manchester	Large	5,043	744	0.15
PopBeauty Coventry	Medium	4,771	723	0.15
PopBeauty Edinburgh	Large	4,615	613	0.13
PopBeauty Bath	Boutique	4,946	593	0.12
PopBeauty Birmingham	Large	4,870	534	0.11
PopBeauty York	Medium	5,360	598	0.11
PopBeauty Canterbury	Boutique	4,994	449	0.09
PopBeauty Exeter	Boutique	4,720	427	0.09
PopBeauty Liverpool	Medium	4,538	442	0.10
PopBeauty Cardiff	Medium	4,540	515	0.11
PopBeauty Leeds	Large	4,305	494	0.12

Table 1: Units on Hand vs. Units Sold and Sell-through Ratio by Store

The London Flagship converts 43% of its stock into sales, far above Glasgow at 16% and the Boutiques at 9-12%, despite holding the fewest units (see Appendix B.1). This shows that high-demand stores are understocked while smaller stores are overstocked. The analysis also identifies 135 below-minimum stock situations and 163 high-risk expiry situations (see Appendix C.1 and C.2).

4. Key Business Insights

4.1. Premium Product Misallocation

High-value Beauty Tools and Fragrance are misallocated across stores. Bath, Canterbury, and Exeter hold large volumes of Dyson Airwrap and FOREO LUNA 4 products, while the Flagship holds only 2–5 units despite critical status (see Appendix E.1). Cumulatively, each of the three boutiques hold around 280-300 premium Beauty Tool units, compared to just 11-16 units in the Flagship's.

Sales data analysis shows that the Flagship is the primary market for these products (see Appendix E.1). It sold 104 units of premium Beauty Tools, while Canterbury sold 1 unit and Exeter sold 0. The Flagship's sell-through ratio for Beauty Tools stands at 0.41, compared to 0.06 for

Canterbury, and 0.02 for Exeter. For Fragrance, the disparity is equally striking. Bath holds 453 Fragrance units and records 0 Fragrance sales, while the Flagship holds 576 units and generates £18,608 in Fragrance revenue at a 0.15 sell-through rate. This pattern is consistent with what Kumar and Vrat (2006) describe as store-profile-blind replenishment, where identical stocking decisions are applied regardless of each location's customer profile.

4.2. *Expiry Risk in Short-Life Skincare*

A second significant finding concerns perishable skincare products in the West Midlands (see Appendix C.4). Birmingham and Coventry hold 344 and 375 units respectively, with the lowest average days-to-expiry in the network (146.6 and 184.8 days). At the same time, both stores show low monthly sales rates (0.3-0.8 units per SKU) for the affected products, suggesting a material proportion will expire before it can be sold. Table 2 shows the top estimated write-off positions.

Store	Product	Units at Risk	Monthly Sales	Est. Write-off
Glasgow	Tatcha Dewy Skin Cream 30ml	115	0.10	£13,800
Birmingham	Glow Recipe Watermelon Dew Drops	80	0.80	£7,189
Coventry	Glow Recipe Watermelon Dew Drops	79	0.30	£7,050
Edinburgh	Tatcha Dewy Skin Cream 30ml	55	0.00	£6,569
Coventry	Tatcha Dewy Skin Cream 30ml	50	0.00	£5,972
Liverpool	Kiehl's Ultra Facial Cream 30ml	83	0.00	£4,884
Coventry	Kiehl's Ultra Facial Cream 30ml	79	0.70	£4,677
Birmingham	Kiehl's Ultra Facial Cream 30ml	76	0.30	£4,486

Table 2: Top Estimated Skincare Write-off Positions by Store and Product

This situation is avoidable because the same products at risk of expiry in Birmingham and Coventry are at critical shortage in Edinburgh and Glasgow. Edinburgh holds just 6 units of Kiehl's Ultra Facial Cream against a reorder threshold of 32, and Glasgow holds only 2 units (see Appendix Table C.3.1). Redistributing stock from the Midlands to Scotland could reduce write-off risk while resolving shortage, without additional procurement cost.

4.3. *North Warehouse Fulfilment Bottleneck*

Manchester and Liverpool stores are in critical shortage for core Makeup products (see Appendix D.3). For example, Manchester holds only 1 unit of Too Faced Foundation (reorder threshold 45), 1 of NARS Concealer (reorder threshold 44), and 5 of Fenty Primer (reorder threshold 38). While

this shows a need for new procurement, warehouse inventory reveals that the North Warehouse in Manchester holds thousands of units of these exact products and the stock exists less than 10 miles from the store.

Further investigation of the Shipments table reveals the root cause (check Appendix D.2). All 25 shipments from the North Warehouse to these stores remain either In Transit or Partially Received. The oldest of these, shipment SHP00860, was dispatched on 25 February 2025 and has remained In Transit for 394 days. Rushton *et al* (2022) claim that standard intra-UK warehouse-to-store delivery is completed within 1-3 business days. Hence, a shipment open for 394 days is not in transit, but a ghost record. Since the inventory management system treats these shipments as pending, it suppresses the generation of fresh replenishment orders, leaving both stores in ongoing shortage.

4.4. Gross Profit by Category and Store Size

The gross profit analysis reveals important patterns about where profitability is concentrated across the network. Table 3 shows the results by category.

Category	Units Sold	Revenue	Gross Profit	Margin
Makeup	5,200	£331,427	£186,432	56%
Fragrance	488	£71,530	£38,499	54%
Skincare	1,319	£71,360	£33,240	47%
Beauty Tools	541	£53,947	£31,643	59%
Haircare	861	£32,150	£14,689	46%
Beauty Accessories	236	£7,454	£3,972	53%

Table 3: Gross Profit and Margin by Product Category

Beauty Tools carries the highest gross margin at 59%, followed by Makeup at 56%. This makes the premium tool misallocation described in Section 4.1 not only a revenue problem but a disproportionate margin problem because the products on which PopBeauty earns the most per unit are the ones sitting idle in boutique stores. At the store level, the Flagship generates £92,010 in gross profit, over 3.5 times Glasgow at £25,292. In contrast, Bath Boutique ranks third at £20,934 but has a sell-through rate for Beauty Tools of 0.07, indicating limited demand (see Appendix B.2 and Appendix Table E.2.1).

4.5. Reorder Level Miscalibration

A structural explanation for these imbalances lies in how reorder levels are configured. The analysis calculates reorder coverage months, which is the number of months of supply the current

reorder threshold represents at each store's actual average monthly sales rate per SKU. Table 4 presents the results.

Store	Avg. Reorder Level	Avg. Monthly Sales/SKU	Coverage (months)
PopBeauty London (Flagship)	29.0	1.21	24
PopBeauty Manchester	26.2	0.43	60
PopBeauty Glasgow	29.4	0.43	69
PopBeauty Edinburgh	28.2	0.36	79
PopBeauty Bath	25.4	0.25	101
PopBeauty Birmingham	29.4	0.22	136
PopBeauty Leeds	29.0	0.15	190
PopBeauty Canterbury	26.7	0.15	182
PopBeauty Exeter	28.2	0.15	194
PopBeauty Liverpool	27.1	0.12	223

Table 4: Reorder Level Calibration by Store

The reorder threshold for the London Flagship represents 24 months of supply at its actual sales rate. In theory, this appears protective. In practice, however, the Flagship's velocity of 1.21 units per SKU per month means the threshold of 29 units is consumed in roughly three and a half weeks, far sooner than the system assumes. In contrast, Liverpool's reorder level of 27 units represents 223 months of supply at its 0.12 monthly rate, implying replenishment is rarely triggered. Therefore, the same numerical threshold produces entirely different operational outcomes depending on store velocity. Silver *et al* (2017) describe this as the uniform policy fallacy in inventory management, where a single policy applied across heterogeneous locations systematically over-replenishes slow stores and under-replenishes fast ones.

4.6. *Discounting as a Symptom Rather Than a Solution*

The final insight concerns pricing behaviour at smaller stores. Bath, York, and Coventry are applying repeated 17-19% discounts to short shelf-life products, yet the inventory remains unsold (see Appendix E.3. and E.4). For example, York still holds 110 units of Too Faced Born This Way Foundation with only 95 days to expiry, despite multiple discount attempts and minimal sales.

Price comparison analysis reveals why discounting is failing. Flagship and Large stores sell the same products without discount, achieve average selling prices 38-45% higher, and average between 6-7 units per transaction compared to 1.9-2.6 at Boutiques and Medium stores. Talluri

and van Ryzin (2004) claim maximum effectiveness of markdown strategies upon the existence of price-sensitive demand. When demand does not exist at a given location, the markdown erodes margin without clearing inventory. In this case, redistribution to a location with genuine demand is the superior intervention.

5. Recommendations

The insights indicate that inventory is being managed at network level rather than store level. The recommendations are grouped by urgency but address root causes rather than symptoms.

5.1. Immediate (2 Weeks)

The most urgent priority is to move premium Beauty Tools from Boutiques to the Flagship. Bath, Canterbury, and Exeter each hold 280-300 units of the Dyson Airwrap and FOREO LUNA 4 at EXCESS status, while the London Flagship holds 2-5 units at CRITICAL. Approximately 150- 200 units should be transferred, prioritising both Dyson and FOREO variants. Since Beauty Tools carry the highest gross margin 59% (see Appendix B.2), this reallocation recovers both revenue and margin.

Short-life Skincare should also be redistributed from Birmingham and Coventry to Edinburgh and Glasgow. Birmingham holds 344 units (averaging 146.6 days to expiry), Coventry 375 units (184.8 days to expiry), with low sell rates of 0.3-0.8 units per SKU per month, a material proportion will expire before it can be sold. The same SKUs are at critical shortage in Edinburgh and Glasgow (see Appendix C.3), making redistribution cost-effective.

Finally, all 25 “In Transit” shipments from the North Warehouse to Manchester and Liverpool should be audited (see Appendix D.2). The oldest record has been open for 394 days. Since inventory system treats these as pending, it blocks replenishment and should be written off and replaced.

5.2. Short-term (4-8 Weeks)

In parallel with the redistribution, discounting at York, Bath, and Coventry should be halted for the three products in Section 4.6. Discounts of 17-19% has not cleared inventory, with expiry in 95-116 days. These units should instead be redirected to the Flagship and Large stores, where they sell at no discount and 3-4 times higher volumes (see Appendix E.3 and E.4).

Additionally, a full network expiry review should cover all 163 cases in Section 3. Each should be assessed against destination sell rates to determine whether redistribution or promotional clearance is appropriate (see Appendix C.2 for the full expiry risk output).

5.3. *Structural (A Quarter)*

The most consequential structural change is the recalibration of reorder levels across all stores using store-specific velocity data. As Section 4.5 and Appendix D.1 demonstrate, the current system applies near-identical thresholds across a network where sell rates vary by a factor of 10. A straightforward formula would be:

$$\text{Reorder Level} = \frac{(\text{Avg. Monthly Sales per SKU}) \times \text{Target Weeks of Cover}}{\text{Avg. number of weeks in a month}}$$

Cover targets should be differentiated by store tier; 8 weeks for the Flagship, 12 for Large stores, 16 for Medium stores, and 20 for Boutiques. Under this formula, the Flagship's threshold for a product selling 1.21 units per month would be approximately 22 units on an 8-week cover, compared to the current equivalent of 24 months of supply.

Alongside reorder recalibration, a maximum "In Transit" threshold of 7 days should be implemented in the warehouse system, after which unresolved shipments trigger an automatic escalation flag. Finally, a quarterly stock rebalancing process should be established as a formal cross-store review of sell-through ratios by category and tier, to identify redistribution opportunities before they become write-off risks.

6. Conclusion

This project demonstrates how structured data highlight issues that would be difficult to identify through qualitative observation. PopBeauty's inventory imbalance results from data and process failure rather than supply shortage. Stock exists, but allocation is not aligned with store-level demand, reorder thresholds ignore store-level velocity, and fulfilment pipeline contains ghost records which block replenishment. Consequently, the highest-revenue store faces shortage while boutiques accumulate excess expiring inventory.

The remedies are largely internal and achievable without additional capital expenditure. Redistributing stock, resolving shipment records, and adjusting reorder thresholds can significantly improve performance.

7. References

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8. Appendices

Appendix A: Database Design

The table below summarises all nine entities in the PopBeauty database, including primary keys, foreign keys, and cardinality relationships.

Entity	Primary Key	Key Foreign Keys	Cardinality
Stores	store_id	None	1:M to Sales, Store Inventory, Shipments
Warehouses	warehouse_id	None	1:M to Warehouse Inventory, Shipments
Brands	brand_id	None	1:M to Products
Categories	category_id	None	1:M to Products
Products	product_id	brand_id, category_id	1:M to Sales, Store Inventory, Warehouse Inventory, Shipments
Store Inventory	inventory_id	store_id, product_id	M:1 to Stores and Products
Warehouse Inventory	warehouse_inventory_id	warehouse_id, product_id	M:1 to Warehouses and Products
Sales	sale_id	store_id, product_id	M:1 to Stores and Products
Shipments	shipment_id	warehouse_id, store_id, product_id	M:1 to Warehouses, Stores, Products

Table A: Entity Summary

Appendix B: Sales and Inventory Overview

B.1. Total Units on Hand by Store

The Flagship holds fewer units than any other store despite generating the highest revenue in the network.

Store	Size	Region	SKUs	Units on Hand	Avg Units/SKU
PopBeauty York	Medium	Yorkshire	84	5,360	63.8
PopBeauty Manchester	Large	North West	92	5,043	54.8
PopBeauty Canterbury	Boutique	South East	78	4,994	64.0
PopBeauty Bath	Boutique	South West	78	4,946	63.4
PopBeauty Birmingham	Large	West Midlands	82	4,870	59.4
PopBeauty Glasgow	Large	Scotland	80	4,832	60.4
PopBeauty Coventry	Medium	West Midlands	83	4,771	57.5
PopBeauty Exeter	Boutique	South West	81	4,720	58.3
PopBeauty Edinburgh	Large	Scotland	76	4,615	60.7
PopBeauty Cardiff	Medium	Wales	73	4,540	62.2
PopBeauty Liverpool	Medium	North West	76	4,538	59.7
PopBeauty Leeds	Large	Yorkshire	72	4,305	59.8
PopBeauty London	Flagship	South East	83	4,075	49.1

Table B.1: Total Units on Hand by Store

B.2. Revenue and Profit Analysis

The Flagship generates £92,010 in gross profit, more than 3.5 times the next highest store.

Store	Units Sold	Revenue	Gross Profit	Margin
PopBeauty London	1,746	£165,658	£92,010	56%
PopBeauty Glasgow	767	£48,960	£25,292	52%
PopBeauty Manchester	744	£45,019	£23,459	52%
PopBeauty Bath	593	£36,447	£20,934	57%
PopBeauty Coventry	723	£37,699	£20,764	55%
PopBeauty Birmingham	534	£34,000	£18,816	55%
PopBeauty York	598	£32,133	£17,992	56%
PopBeauty Cardiff	515	£31,530	£16,982	54%
PopBeauty Edinburgh	613	£35,308	£16,595	47%
PopBeauty Leeds	494	£28,568	£15,932	56%
PopBeauty Exeter	427	£25,403	£14,040	55%
PopBeauty Canterbury	449	£25,059	£13,535	54%
PopBeauty Liverpool	442	£22,084	£12,125	55%

Table B.2.1: Gross Profit by Store

Boutiques generate only £10,110 in Fragrance revenue combined, compared to £22,432 at the Flagship alone.

Tier	Category	Units Sold	Revenue
Flagship	Makeup	1,328	£109,355
Flagship	Beauty Tools	116	£24,126
Flagship	Fragrance	116	£22,432
Flagship	Haircare	135	£6,371
Flagship	Skincare	51	£3,374
Flagship	Beauty Accessories	0	£0
Large	Makeup	1,686	£96,566
Large	Skincare	708	£43,002
Large	Fragrance	148	£23,407
Large	Beauty Tools	184	£12,804
Large	Haircare	302	£12,083
Large	Beauty Accessories	124	£3,992

Medium	Makeup	1,292	£73,162
Medium	Skincare	348	£15,701
Medium	Fragrance	156	£15,581
Medium	Haircare	285	£8,908
Medium	Beauty Tools	139	£8,191
Medium	Beauty Accessories	58	£1,904
Boutique	Beauty Accessories	54	£109,355
Boutique	Makeup	894	£52,344
Boutique	Fragrance	68	£10,110
Boutique	Skincare	212	£9,284
Boutique	Beauty Tools	102	£8,826
Boutique	Haircare	139	£4,788

Table B.2.2: Revenue by Category and Store Tier

Appendix C: Expiry Risk and Inventory Health

C.1. Stores Below Minimum Stock Level

There are 135 shortage situations across the network. The 15 most severe are shown below, sorted by units below minimum, but for the full list, please review Section C.1. in the code submitted.

Store	Category	Product	Qty	Min	Deficit
PopBeauty Canterbury	Makeup	Benefit POREfessional Primer 30ml	3	37	-34
PopBeauty Edinburgh	Beauty Tools	Real Techniques Expert Face Brush Black	3	37	-34
PopBeauty Leeds	Haircare	ghd Bodyguard Heat Protect Spray 150ml	7	40	-33
PopBeauty Exeter	Makeup	Clinique Even Better Serum Foundation	4	34	-30
PopBeauty Coventry	Makeup	MAC Prep + Prime Fix+ Translucent	7	35	-28
PopBeauty London	Makeup	NARS Radiant Creamy Concealer Tan 1	6	34	-28
PopBeauty London	Haircare	Living Proof Clean Dry Shampoo 200ml	14	42	-28
PopBeauty Coventry	Fragrance	Prada Paradoxe Eau de Parfum Duo Set	8	35	-27
PopBeauty Glasgow	Beauty Tools	Beautyblender Original	1	28	-27
PopBeauty Leeds	Makeup	Urban Decay All Nighter Setting Spray	6	33	-27
PopBeauty London	Skincare	NUXE Huile Prodigieuse Dry Oil 50ml	10	37	-27
PopBeauty London	Makeup	Rare Beauty Brightening Concealer Med 1	2	28	-26
PopBeauty Cardiff	Makeup	MAC Prep + Prime Natural Radiance 30ml	7	31	-24
PopBeauty Birmingham	Makeup	NARS Afterglow Liquid Blush Coral Crush	17	40	-23
PopBeauty Exeter	Accessories	Slip Travel Vanity Mirror Black	13	35	-22

Table C.1: Top 15 Stockout Situations by Severity

C.2. High-Risk Expiry Situations

There are 163 product-store combinations where more than 30 units are held with fewer than 6 months to expiry (calculations are anchored to March 26th, 2026). The 15 most time-critical are shown below, but please review Section C.3. in the code for the extended output.

Store	Product	Category	Qty	Expiry	Days Left
PopBeauty York	Laneige Water Sleeping Mask 75ml	Skincare	83	2026-04-15	20
PopBeauty Bath	Benefit Benetint Hope	Makeup	46	2026-04-16	21
PopBeauty Cardiff	Moroccanoil Color Depositing Mask 150ml	Haircare	94	2026-04-18	23
PopBeauty Liverpool	Tweezerman Slant Tweezer White	Beauty Tools	75	2026-04-18	23
PopBeauty Edinburgh	MAC Prep + Prime Fix+ Translucent	Makeup	73	2026-04-18	23
PopBeauty Edinburgh	The Ordinary Hyaluronic Acid 2%+B5 50ml	Skincare	54	2026-04-20	25
PopBeauty Manchester	Rare Beauty Soft Pinch Blush Petal	Makeup	90	2026-04-23	28
PopBeauty Manchester	Jo Malone English Pear Cologne 50ml	Fragrance	91	2026-04-25	30
PopBeauty York	Lancôme Teint Idole Foundation 210N	Makeup	61	2026-04-25	30
PopBeauty Birmingham	Calvin Klein CK One Eau de Toilette	Fragrance	73	2026-04-26	31
PopBeauty Coventry	Dior Sauvage Eau de Parfum 50ml	Fragrance	89	2026-04-28	33
PopBeauty Coventry	Benefit Hoola Matte Bronzer Medium	Makeup	38	2026-04-28	33
PopBeauty Canterbury	Charlotte Tilbury Matte Lipstick Ruby	Makeup	93	2026-04-29	34
PopBeauty Edinburgh	Laneige Lip Sleeping Mask 20g	Skincare	116	2026-04-30	35
PopBeauty Bath	Paula's Choice Skin Perfecting 2% BHA Liquid Exfoliant 100ml	Skincare	102	2026-05-03	38

Table C.2: Top 15 High-risk Expiry Situations

C.3. Scotland Critical Shortage

Edinburgh and Glasgow are at critical shortage on the exact SKUs expiring unsold in Birmingham and Coventry.

Store	Product	Stock	Min Level	Reorder Level	Status
PopBeauty Edinburgh	Glow Recipe Watermelon Dew Drops 30ml	2	15	40	CRITICAL SHORTAGE
PopBeauty Glasgow	Glow Recipe Watermelon Dew Drops 30ml	5	17	39	CRITICAL SHORTAGE
PopBeauty Glasgow	Kiehl's Ultra Facial Cream 30ml	2	18	31	CRITICAL SHORTAGE
PopBeauty Edinburgh	Kiehl's Ultra Facial Cream 30ml	6	15	32	CRITICAL SHORTAGE
PopBeauty Glasgow	La Roche-Posay Effaclar Foaming Gel 400ml	6	20	36	CRITICAL SHORTAGE
PopBeauty Edinburgh	La Roche-Posay Effaclar Foaming Gel 400ml	7	13	33	CRITICAL SHORTAGE
PopBeauty Glasgow	Supergoop! Unseen Sunscreen SPF30 50ml	3	14	33	CRITICAL SHORTAGE
PopBeauty Edinburgh	Supergoop! Unseen Sunscreen SPF30 50ml	8	12	35	CRITICAL SHORTAGE

Table C.3: Scotland Critical Shortage

C.4. Short Shelf-Life Skincare by Store

Birmingham and Coventry hold the largest volumes of short shelf-life Skincare at the shortest average days to expiry in the network.

Store	Size	Region	Short-life SKUs	Total Units	Min Days	Avg Days
PopBeauty Birmingham	Large	West Midlands	5	344	113	147.6
PopBeauty Bath	Boutique	South West	5	422	38	183.8
PopBeauty Coventry	Medium	West Midlands	5	375	146	184.8
PopBeauty Canterbury	Boutique	South East	5	352	73	229.8
PopBeauty York	Medium	Yorkshire	4	243	172	234.3
PopBeauty Exeter	Boutique	South West	6	257	115	248.8
PopBeauty Liverpool	Medium	North West	4	243	125	261.8
PopBeauty London	Flagship	South East	5	301	94	267.2
PopBeauty Edinburgh	Large	Scotland	7	311	137	277.1
PopBeauty Cardiff	Medium	Wales	4	330	234	280.0
PopBeauty Manchester	Large	North West	5	231	149	303.6
PopBeauty Leeds	Large	Yorkshire	5	279	191	305.6
PopBeauty Glasgow	Large	Scotland	6	160	294	321.0

Table C.4: Short shelf-life Skincare by store (sorted by average days to expiry)

Appendix D: Warehouse and Reorder Analysis

D.1. Reorder Level Calibration

The coverage column shows how many months of supply the current reorder level represents at each store's actual average sales rate.

Store	Size	Avg Reorder Level	Avg Monthly Sales/SKU	Coverage (months)
PopBeauty London	Flagship	29.0	1.21	24
PopBeauty Manchester	Large	26.2	0.43	60
PopBeauty Glasgow	Large	29.4	0.43	69
PopBeauty Edinburgh	Large	28.2	0.36	79
PopBeauty Coventry	Medium	27.8	0.30	94
PopBeauty York	Medium	27.7	0.29	96
PopBeauty Bath	Boutique	25.4	0.25	101
PopBeauty Birmingham	Large	29.4	0.22	136
PopBeauty Cardiff	Medium	28.8	0.20	145
PopBeauty Leeds	Large	29.0	0.15	190
PopBeauty Canterbury	Boutique	26.7	0.15	182
PopBeauty Exeter	Boutique	28.2	0.15	194
PopBeauty Liverpool	Medium	27.1	0.12	223

Table D.1: Reorder Level Calibration

D.2. Stuck Shipments from North Warehouse

All 25 shipments from the North Warehouse to the stores in Manchester and Liverpool remain unresolved. The 10 oldest are shown below. The extended output can be found on Section G.2. of the code (anchored to March 26th 2026).

Shipment ID	Destination	Product	Qty	Dispatched	Status	Days
SHP00860	PopBeauty Manchester	Moroccanoil Color Depositing Mask 150ml	110	2025-02-25	In Transit	394
SHP00414	PopBeauty Liverpool	Laneige Water Sleeping Mask 75ml	29	2025-07-16	In Transit	253
SHP00979	PopBeauty Manchester	Dior Sauvage Eau de Parfum 50ml	42	2025-11-20	In Transit	126
SHP01007	PopBeauty Liverpool	Charlotte Tilbury Flawless Foundation 260N	92	2026-01-27	In Transit	58
SHP01009	PopBeauty Manchester	NARS Radiant Creamy Concealer Medium 2	86	2026-01-27	In Transit	58
SHP01001	PopBeauty Manchester	Fenty Beauty Pro Filt'r Foundation 110N	98	2026-01-29	In Transit	56
SHP01017	PopBeauty Manchester	Fenty Beauty Pro Filt'r Primer 50ml	97	2026-01-31	In Transit	54
SHP01005	PopBeauty Manchester	Charlotte Tilbury Flawless Foundation 260N	98	2026-02-01	In Transit	53
SHP01011	PopBeauty Liverpool	NARS Radiant Creamy Concealer Medium 2	82	2026-02-02	In Transit	52
SHP01019	PopBeauty Liverpool	Fenty Beauty Pro Filt'r Primer 50ml	101	2026-02-07	In Transit	47

Table D.2: Top 10 Stuck Shipments from the North Warehouse

D.3 North Warehouse Stock vs. Store Shortages

The North Warehouse (WH002) in Manchester holds thousands of units of the exact products that Manchester and Liverpool are critically short on. The following table displays only 10 of these, but the full output can be found on Section G.1. of the code.

Product	Category	Warehouse Stock	Store	Store Stock	Reorder	Status
The Ordinary Hyaluronic Acid 2%+B5 15ml	Skincare	5,741	PopBeauty Manchester	7	12	BELOW REORDER
Charlotte Tilbury Concealer Medium 2	Makeup	5,554	PopBeauty Liverpool	11	40	BELOW REORDER
Too Faced Foundation 330N	Makeup	5,198	PopBeauty Manchester	1	45	CRITICAL
Tweezerman Slant Tweezer Rose Gold	Beauty Tools	4,957	PopBeauty Liverpool	3	15	CRITICAL
Rare Beauty Soft Pinch Blush Hope	Makeup	4,671	PopBeauty Manchester	6	32	CRITICAL
Rare Beauty Soft Pinch Blush Hope	Makeup	4,671	PopBeauty Liverpool	7	39	CRITICAL
Slip Silk Scrunchies Champagne	Beauty Accessories	4,661	PopBeauty Liverpool	29	37	BELOW REORDER
MAC Studio Fix Foundation 110N	Makeup	4,658	PopBeauty Manchester	3	37	CRITICAL
NARS Radiant Creamy Concealer Medium 2	Makeup	4,281	PopBeauty Manchester	1	44	CRITICAL
NARS Radiant Creamy Concealer Medium 2	Makeup	4,281	PopBeauty Liverpool	3	44	CRITICAL

Table D.3: North Warehouse Stock vs. Store Shortages

Appendix E: Premium Products and Discount Analysis

E.1. Premium Beauty Tools: Stock Status by Store

EXCESS status is concentrated at Boutique and Medium stores; CRITICAL at the Flagship. The extended output can be found in Section E.1. of the code.

Store	Size	Product	Price	Stock	Reorder	Status
Cardiff	Medium	Dyson Airwrap Black	£315.71	119	26	EXCESS
Canterbury	Boutique	Dyson Airwrap Black	£315.71	100	18	EXCESS
Bath	Boutique	Dyson Airwrap Black	£315.71	92	15	EXCESS
Exeter	Boutique	Dyson Airwrap Black	£315.71	91	25	EXCESS
Liverpool	Medium	Dyson Airwrap Black	£315.71	16	13	OK
London	Flagship	Dyson Airwrap Black	£315.71	2	42	CRITICAL
Coventry	Medium	Dyson Airwrap Pink	£328.26	104	38	OK
Canterbury	Boutique	Dyson Airwrap Pink	£328.26	99	22	EXCESS
Bath	Boutique	Dyson Airwrap Pink	£328.26	97	25	EXCESS
Exeter	Boutique	Dyson Airwrap Pink	£328.26	85	17	EXCESS
York	Medium	Dyson Airwrap Pink	£328.26	8	45	CRITICAL
London	Flagship	Dyson Airwrap Pink	£328.26	4	41	CRITICAL
Exeter	Boutique	FOREO LUNA 4 Mint	£284.73	117	24	EXCESS
Bath	Boutique	FOREO LUNA 4 Mint	£284.73	112	15	EXCESS
Canterbury	Boutique	FOREO LUNA 4 Mint	£284.73	90	18	EXCESS
London	Flagship	FOREO LUNA 4 Mint	£284.73	5	41	CRITICAL

Table E.1.1: Dyson Airwrap and FOREO LUNA 4 Stock Status by Store

The Flagship has sold 104 units of Dyson Airwrap and FOREO products across the analysis period, which is more than double the next store (Bath at 37, partly driven by tourist footfall). Canterbury has sold 1 unit. Exeter has sold 0 of these products despite holding substantial inventory.

Store	Category	Units Sold
London	Flagship	104
Bath	Boutique	37
Birmingham	Large	20
Coventry	Medium	17
Leeds	Large	8
Cardiff	Medium	7
Edinburgh	Large	7

Manchester	Large	6
Liverpool	Medium	2
Canterbury	Boutique	1

Table E.1.2: Premium Beauty Tool Sales by Store

E.2. Beauty Tools and Fragrance Sell-Through by Store

The Flagship converts 41% of Beauty Tools stock into sales. Bath holds 453 Fragrance units and records 0 Fragrance sales.

Store	Size	Category	Stock	Sold	Revenue	Sell-Through
London	Flagship	Beauty Tools	263	109	£24,041	0.41
Edinburgh	Large	Beauty Tools	66	9	£119	0.14
York	Medium	Beauty Tools	298	36	£347	0.12
Glasgow	Large	Beauty Tools	207	19	£203	0.09
Bath	Boutique	Beauty Tools	576	40	£7,908	0.07
Manchester	Large	Beauty Tools	200	12	£143	0.06
Canterbury	Boutique	Beauty Tools	371	22	£386	0.06
Birmingham	Large	Beauty Tools	328	16	£3,244	0.05
Coventry	Medium	Beauty Tools	521	22	£1,631	0.04
Cardiff	Medium	Beauty Tools	333	12	£2,308	0.04
Leeds	Large	Beauty Tools	294	10	£125	0.03
Exeter	Boutique	Beauty Tools	551	11	£154	0.02
Liverpool	Medium	Beauty Tools	198	0	£0	0.00
London	Flagship	Fragrance	576	88	£18,609	0.15
Cardiff	Medium	Fragrance	557	20	£3,500	0.04
Canterbury	Boutique	Fragrance	640	17	£2,789	0.03
Manchester	Large	Fragrance	427	11	£2,033	0.03
Glasgow	Large	Fragrance	352	8	£1,373	0.02
Birmingham	Large	Fragrance	490	8	£1,266	0.02
Coventry	Medium	Fragrance	382	5	£853	0.01
Edinburgh	Large	Fragrance	422	5	£388	0.01
York	Medium	Fragrance	458	5	£657	0.01
Exeter	Boutique	Fragrance	741	8	£987	0.01
Liverpool	Medium	Fragrance	528	4	£643	0.01
Leeds	Large	Fragrance	315	0	£0	0.00
Bath	Boutique	Fragrance	453	0	£0	0.00

Table E.2: Beauty Tools and Fragrance Sell-through by Store

E.3. Price Comparison: Discounted vs. Full-Price by Store Tier

The Flagship sells the same products at 38 to 45% higher prices and with six to seven times more units per transaction, at zero discount.

Tier	Product	Avg Discount	Avg Price	Avg Units/Transaction	Transactions
Boutique	Moroccanoil Color Depositing Mask	19%	£42.53	1.89	9
Flagship	Moroccanoil Color Depositing Mask	0%	£66.31	6.83	12
Large	Moroccanoil Color Depositing Mask	0%	£64.04	6.08	12
Medium	Moroccanoil Color Depositing Mask	17%	£51.40	2.14	14
Boutique	MAC Studio Fix Foundation 110N	19%	£40.98	2.25	4
Flagship	MAC Studio Fix Foundation 110N	0%	£66.09	7.17	12
Large	MAC Studio Fix Foundation 110N	0%	£71.38	6.42	12
Medium	MAC Studio Fix Foundation 110N	17%	£54.57	2.53	15
Boutique	Too Faced Foundation 330N	17%	£47.01	2.60	5
Flagship	Too Faced Foundation 330N	0%	£68.77	6.75	12
Large	Too Faced Foundation 330N	0%	£63.52	6.58	12
Medium	Too Faced Foundation 330N	17%	£50.91	2.08	12

Table E.3: Price and Transaction Comparison by Store Tier for 3 Dead-stock Products

E.4. Dead Stock: High Inventory with Near Expiry

These products are being discounted repeatedly but stock is not clearing. All expire within 116 days.

Store	Size	Product	Shelf Life	Stock	Expiry	Days Left	Discount	Sold
PopBeauty York	Medium	Too Faced Foundation 330N	14 months	110	2026-06-29	95	18%	7
PopBeauty Bath	Boutique	Moroccanoil Mask 150ml	16 months	105	2026-07-03	99	19%	17
PopBeauty Coventry	Medium	MAC Studio Fix Foundation	16 months	96	2026-07-20	116	17%	19
PopBeauty York	Medium	MAC Studio Fix Foundation	16 months	94	2026-07-20	116	17%	19
PopBeauty Bath	Boutique	Too Faced Foundation 330N	14 months	90	2026-06-29	95	17%	13
PopBeauty York	Medium	Moroccanoil Mask 150ml	16 months	89	2026-07-03	99	17%	14
PopBeauty Coventry	Medium	Too Faced Foundation 330N	14 months	85	2026-06-29	95	17%	18
PopBeauty Coventry	Medium	Moroccanoil Mask 150ml	16 months	83	2026-07-03	99	17%	16
PopBeauty Bath	Boutique	MAC Studio Fix Foundation	16 months	78	2026-07-20	116	19%	9

Table E.4: Dead Stock Positions

9. Task Allocations/ Contributions

Student ID	Contributions
5744242	Propose the business idea, develop business questions, design ER Diagram, record meeting minutes, assist on the report to bring it down within the permitted word count range
5750536	Part 1 (company (locations, warehouses etc.), business problem, KPI's) and made the presentation and co-presented
5745856	Built and updated the database. Did a thorough data analysis and wrote SQL queries. Wrote the report. Flagged and rectified inconsistencies. Made the presentation. Presented.
5754848	Guided the team in ideation and generated business questions, Updated the database and generated the Synthetic Data, Supported the presentation deck preparation process
5697948	Company business ideas, develop business questions. Create and modify the database
5732031	Company selection and business idea, develop of business problem, design of ER diagram, Flagged and rectified inconsistencies.